

Types of Cloud Computing

- **Based on Deployment:**

- 1) Public**
- 2) Private**
- 3) Hybrid**

Based on Services

- 1) SaaS**
- 2) PaaS**
- 3) IaaS**

Public Cloud

- A public cloud is where resources such as storage, servers, and applications are made available to the public over the internet by a third-party provider.
- In a public cloud, the infrastructure is owned and managed by the cloud provider and shared among multiple users or organizations, making it highly scalable and cost-effective.
- Real-Life Examples of Public Cloud Usage: Email Services (e.g., Gmail, Yahoo Mail), Storage Services (e.g., Google Drive, Dropbox, Microsoft OneDrive), Web Hosting (e.g., AWS, Microsoft Azure, Google Cloud Platform), Streaming Services (e.g., Netflix, Spotify)

Private Cloud

- **A private cloud is a cloud computing environment used exclusively by one organization.**
- **The organization may run its private cloud or Outsource it to a hosting company like VMware, Azure etc.**
- **All the services and infrastructure is maintained on a private network.**

<u>Feature</u>	<u>Public Cloud</u>	<u>Private Cloud</u>
Ownership	Third-party provider	Single organization
Infrastructure	Shared (multi-tenant)	Dedicated (single-tenant)
Cost	Lower; pay-as-you-go	Higher; dedicated resources
Scalability	Highly scalable	Scalable but requires planning
Security	Standard security by provider	Enhanced control for sensitive data
Compliance	Limited for regulated industries	Easier to meet regulatory requirements
Customization	Limited	Highly customizable
Maintenance	Managed by provider	Managed by organization
Deployment Speed	Fast, ready-to-use	Slower due to setup requirements
Examples	AWS, Azure, Google Cloud	VMware vSphere, Azure Stack